

1.2: Domain and Range

Definition. (Interval notation)

We can specify the domain and/or range of a function using interval notation:

- Strict inequalities (exclusive) are represented using parenthesis:

$$x > 3 \iff (3, \infty) \qquad x < 9 \iff (-\infty, 9)$$

Note: Parenthesis are **always** used when an interval's end points are infinite

- Nonstrict inequalities (inclusive) are represented using square brackets:

$$7 \leq x \leq 11 \iff [7, 11]$$

- Disjoint intervals can be unioned together:

$$x \neq -2 \iff (-\infty, -2) \cup (-2, \infty)$$

- Any combination of the above can be used:

$$[-1, 1] \cup (2, 3] \cup (4, 5) \cup [6, \infty)$$



Recall that the domain is the set of all *possible* input values.

Example. Find the domain of the following functions (use interval notation): [Graphs](#)

$$f(x) = x - 3$$

$$A = \pi r^2$$

$$y = \sqrt{x - 1}$$

$$y = \frac{x + 2}{x^2 - 4}$$

$$y = \sqrt[3]{x - 4}$$

$$f(x) = \frac{\sqrt[4]{x^2 + 4}}{(x + 1)\sqrt{x - 1}}$$

Example. Identify the domain and range of the following:

[Graphs](#)

$$x(x+1)(x+2)^2, \quad x \geq -2$$

$$\sin(x)$$

$$\log(x)$$

$$2^x, \quad x \geq -1$$

Definition.

A **piecewise** function is a function with different definitions for different portions of the domain.

Example. Rewrite the following as piecewise functions:

$$|x| = \qquad \qquad \qquad \frac{x}{|x|} =$$

$$|x - 1| + |4 - x| =$$

Example. Graph the following piecewise function:

[Graphs](#)

$$f(x) = \begin{cases} x^2, & \text{if } x \leq 1 \\ 3, & \text{if } 1 < x \leq 2 \\ x, & \text{if } x > 2 \end{cases}$$